DATA ANALYTICS ASSIGNMENT - 1

Supermarket Sales Analysis

Historical Sales Data Analysis Across 3 Branches

1. INTRODUCTION

The growth of supermarkets in most populated cities is increasing and market competitions are also high. This analysis examines the historical sales data of a supermarket company which has been recorded across 3 different branches over a 3-month period (January 2019 to March 2019).

Dataset Overview

Data Period: January 2019 - March 2019
Number of Branches: 3 (Branch A, B, and C)
Total Records: 1000 transactions
Product Categories: 6 product lines

2. DATA ATTRIBUTES

The dataset contains the following attributes:

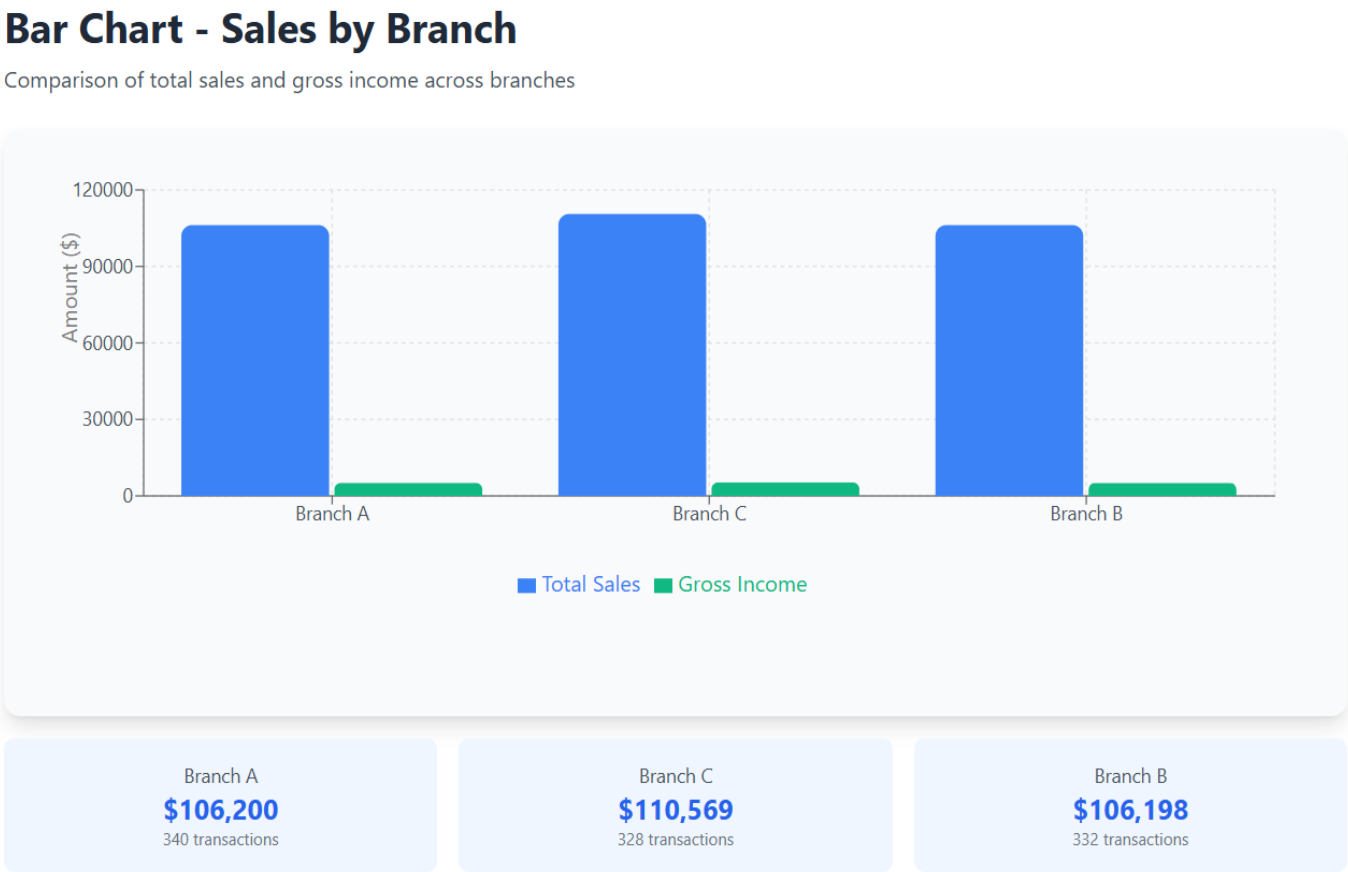
Invoice ID: Computer-generated sales slip invoice identification number
Branch: Branch of supercenter (3 branches: A, B, and C)
City: Location of supercenters
Customer Type: Members (with member cards) or Normal (without member cards)
Gender: Gender type of customer
Product Line: Electronic accessories, Fashion accessories, Food and beverages, Health and beauty, Home and lifestyle, Sports and travel
Unit Price: The price of each product in USD ($)
Quantity: Number of products purchased by the customer
Tax: 5% tax fee for customer purchases
Total: Total price including tax
Date: Date of purchase (January 2019 to March 2019)
Time: Purchase time (10 AM to 9 PM)
Payment: Payment method (Cash, Credit card, or Ewallet)
COGS: Cost of goods sold
Gross Margin Percentage: Gross margin percentage
Gross Income: Gross income from the transaction
Rating: Customer satisfaction rating (scale of 1 to 10)

3. DATA VISUALIZATION

The following visualizations provide insights into various aspects of the supermarket sales data:

3.1 Bar Chart - Sales by Branch

This bar chart compares the total sales and gross income across the three branches (A, B, and C).



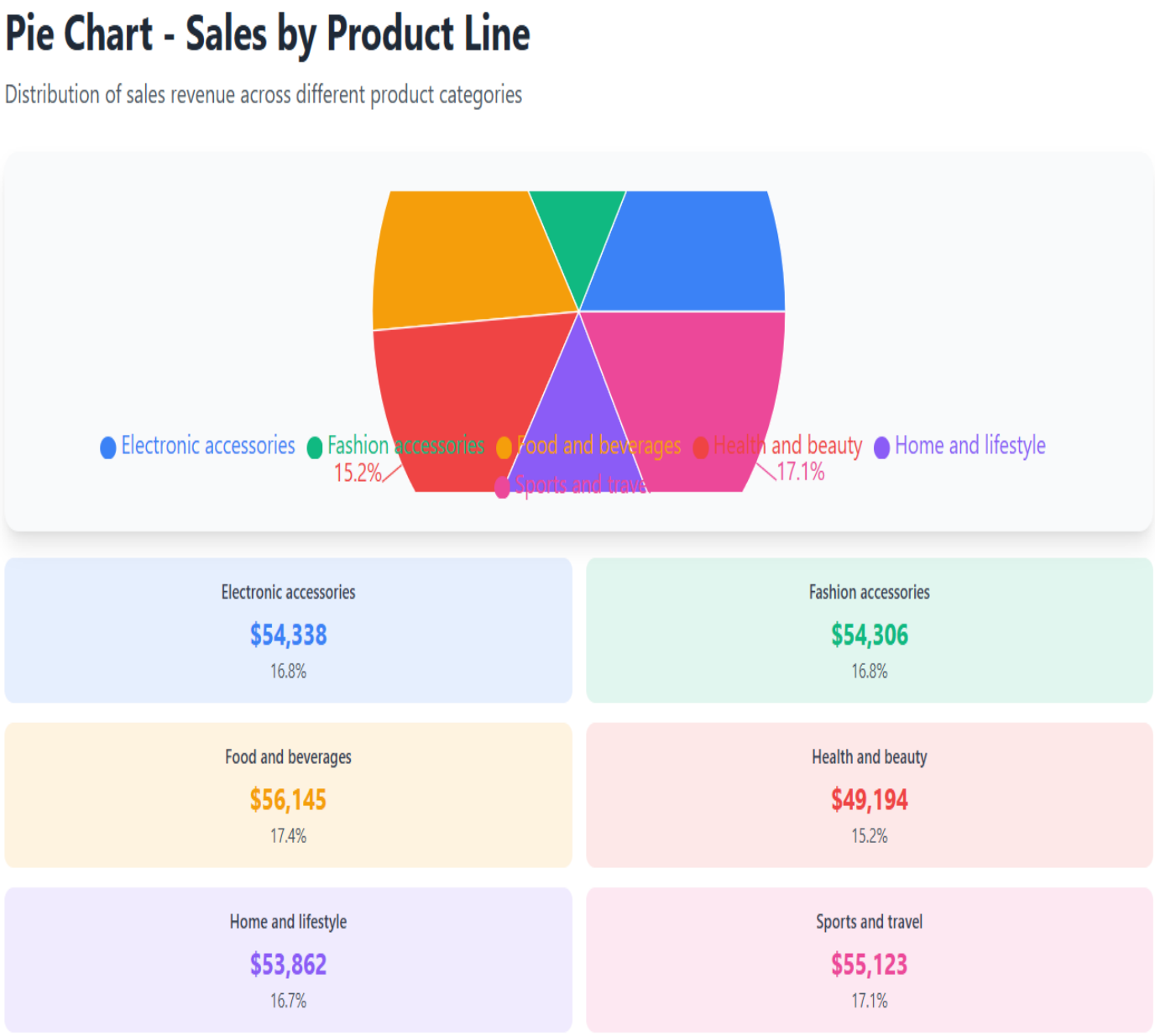
Branch A	Branch C	Branch B
$106,200	$110,569	$106,198
340 transactions	328 transactions	332 transactions

Key Insights:

- Branch performance comparison based on total sales revenue
- Gross income analysis across different locations
- Identification of highest and lowest performing branches

3.2 Pie Chart - Sales by Product Line

This pie chart shows the distribution of sales revenue across different product categories.

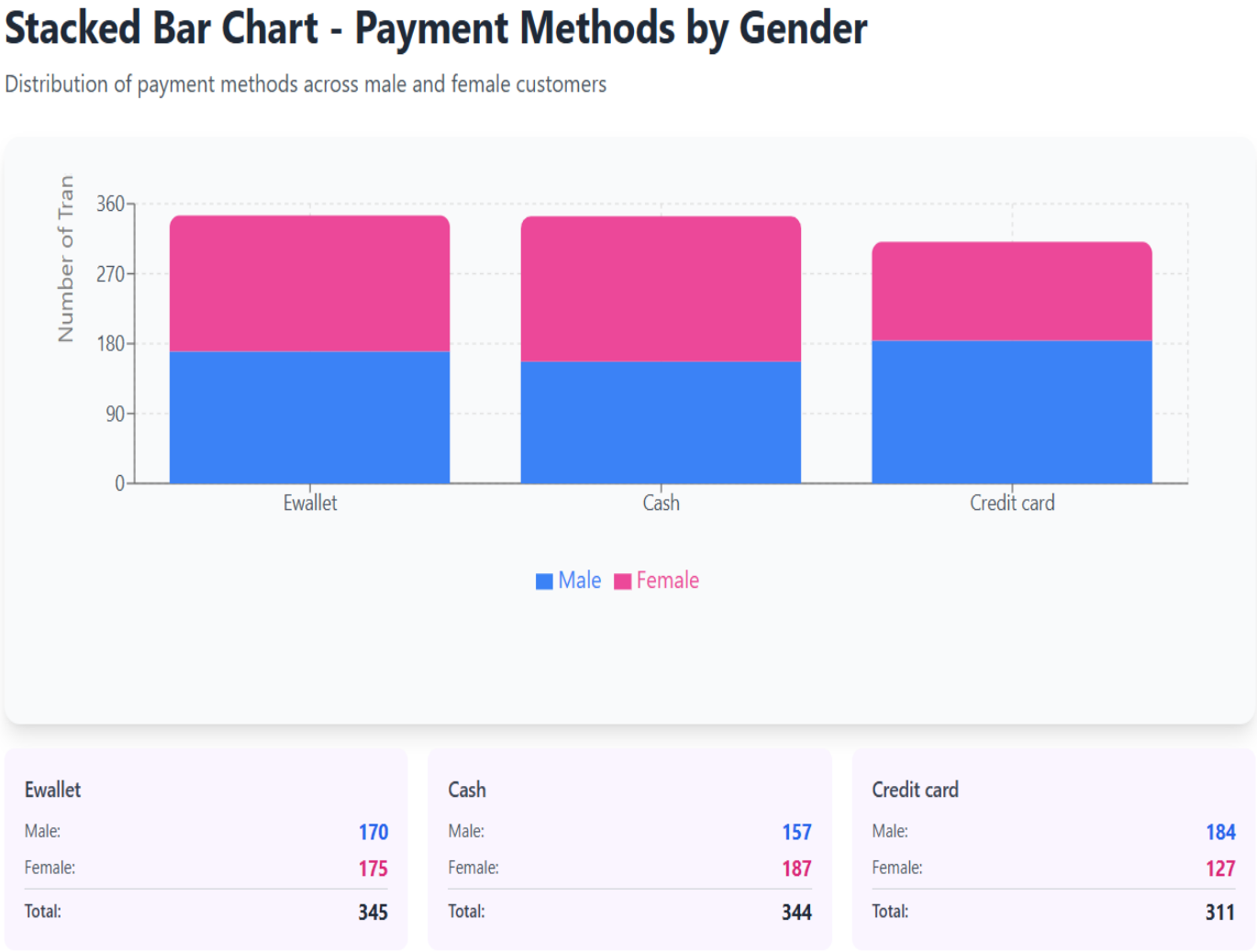


Key Insights:

- Revenue contribution of each product category
- Best-selling product lines in the supermarket
- Product mix analysis for inventory management

3.3 Stacked Bar Chart - Payment Methods by Gender

This stacked bar chart displays the payment method preferences segmented by customer gender.



Ewallet	
Male:	170
Female:	175
Total:	345

Cash	
Male:	157
Female:	187
Total:	344

Credit card	
Male:	184
Female:	127
Total:	311

Key Insights:

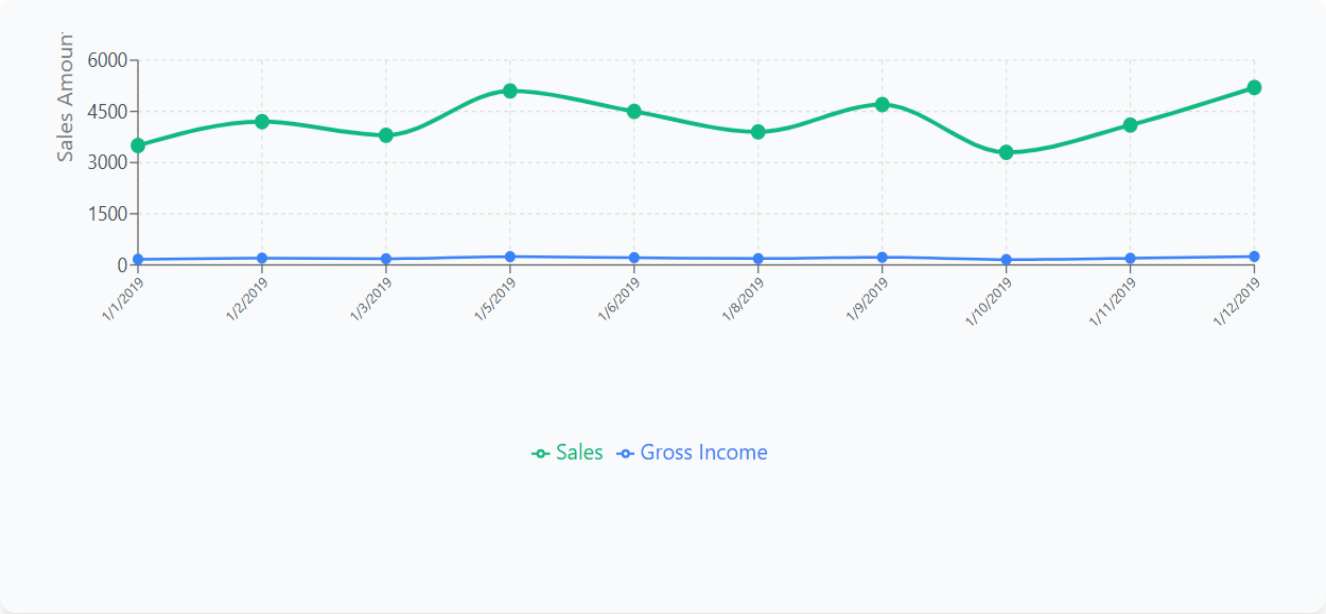
- Payment preference patterns across different genders
- Most popular payment methods (Cash, Credit card, Ewallet)
- Gender-based customer behavior analysis

3.4 Line Chart - Daily Sales Trend

This line chart illustrates the daily sales trends over the three-month period.

Line Chart - Daily Sales Trend

Sales performance over time (January - March 2019)



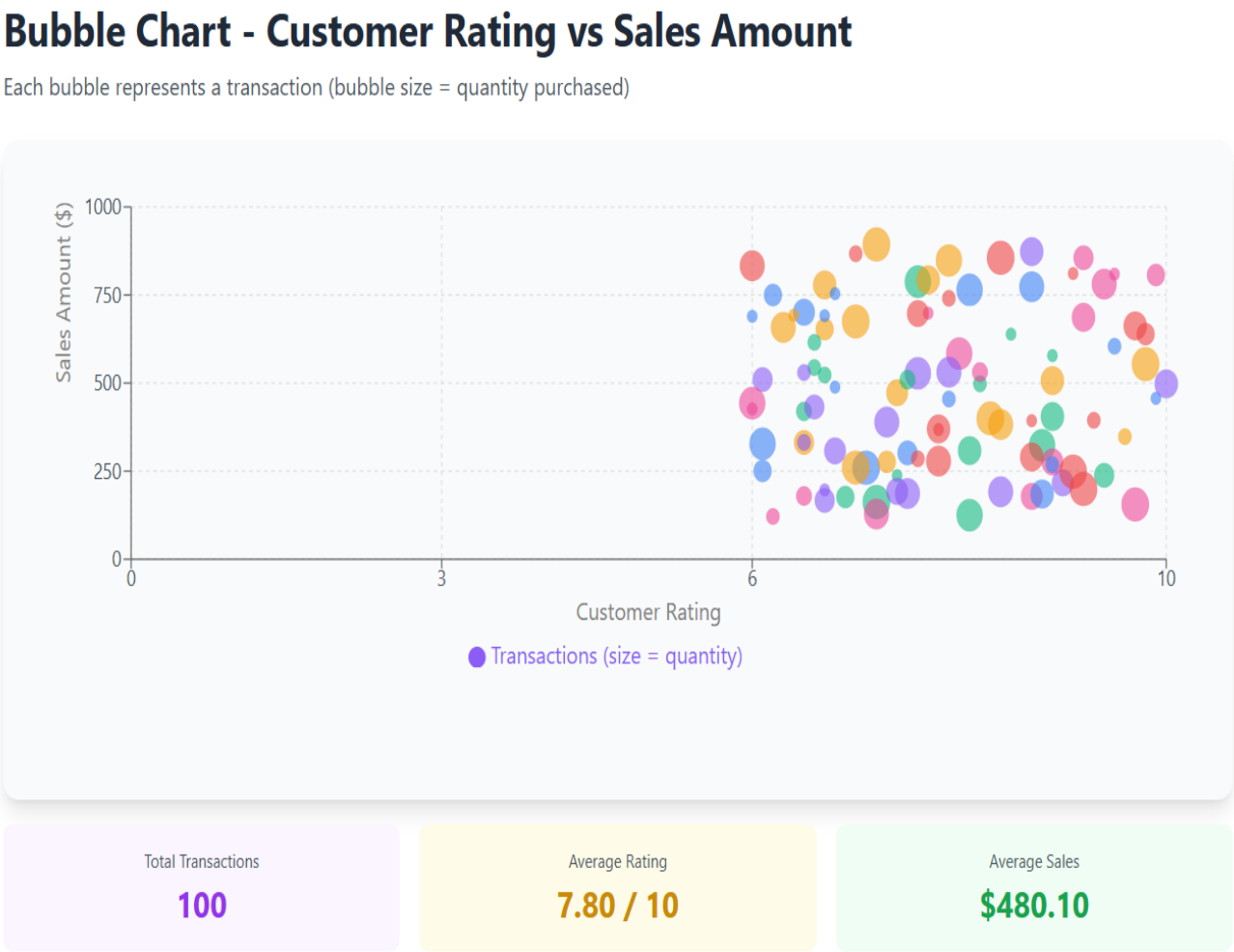
Total Sales	Avg Daily Sales	Highest Day	Lowest Day
$42,300	$4,230	$5,200	$3,300

Key Insights:

- Sales performance trends over time
- Identification of peak and low sales periods
- Seasonal patterns and fluctuations in daily revenue

3.5 Bubble Chart - Customer Rating vs Sales Amount

This bubble chart explores the relationship between customer satisfaction ratings and sales amounts, with bubble size representing quantity purchased.



Total Transactions	Average Rating	Average Sales
100	7.80 / 10	$480.10

Key Insights:

- Correlation between customer satisfaction and purchase amount
- Transaction value distribution across different rating levels
- Impact of quantity purchased on overall sales performance

4. CONCLUSION:

This comprehensive analysis of the supermarket sales data provides valuable insights into various aspects of business performance across three branches. The visualizations reveal important patterns in:

- **Branch Performance:** Comparative analysis of sales and profitability
- **Product Categories:** Revenue distribution across different product lines
- **Customer Behavior:** Payment preferences and gender-based patterns
- **Temporal Trends:** Daily sales fluctuations and patterns
- **Customer Satisfaction:** Relationship between ratings and purchase behavior

These insights can help the supermarket management make data-driven decisions regarding inventory management, marketing strategies, resource allocation, and customer satisfaction improvements.

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